Milkweed Delights

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The Importance of Milkweed

To educate people on the importance of milkweed in its environment and change the perception that milkweed should be eradicated because it is a weed, this cookbook encourages the safe cultivation of milkweed in urban regions by promoting its recipes to create a symbiotic relationship between human beings and milkweed.

Common milkweed is classified as a Class Number One Noxious Weed in Nova Scotia and is prohibited from plantations. The rules and laws are applied to this strand of milkweed in terms of its plantation. Some native milkweeds are driven to extinction because of their believed toxins and the speed they spread in a colonized-like manner,taking the space and nutrients from other crops and having seemingly so little advantage compared to its disadvantages and compared to other plants. (Canadian wildlife federation)

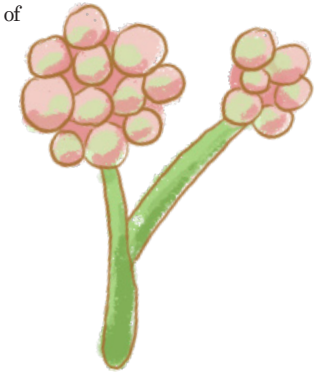
We believe that planting milkweed in a more safe and suitable environment can make its benefits overcome its drawbacks.

What is Milkweed?

Milkweed is the common name for perennial, herbaceous plants of the genus Asclepias and family Asclepiadaceae (Cavers). It is named after the legendary Greek physician and deity Asklepios as it has many uses in medicine (Austin).

There are around 100 species of milkweed worldwide but 13 are native to Canada. Except for butterfly milkweed, all Canadian species are toxic. Six of these species are classified as "weeds" with the most important one, the common milkweed (A. syriaca) being found from regions in Manitoba to the Maritimes (Cavers).

Milkweed is well-known to be the exclusive food source of monarch caterpillars as well as some other insect species and it is also a source of nectar for many insects (Doonan).



From pollination to fruit-bearing

Milkweed flowers are extremely complex as the male and female reproductive parts are attached and both located within the corona. Different from the orange-yellow pollen grains attached to anthers, their pollen is only accessible by going inside the hoods and removing their legs, the protruding horn, thus when the insect travels to another flower, it pollinates it (Canadian Wildlife Federation).

The pollination can only be done by larger insects as the protruding horn needs to be inserted into the slit of another flower. Smaller insects usually die performing the task as they are unable to pull themselves free from the hoods (Canadian Wildlife Federation).

Milkweed fruits are follicles, dry dehiscent fruits, covered with white fiber hair and soft spikes. They can measure up to 12 to 32 centimeters long. Inside the narrow teardrop fruit pods, there lies the 8-millimeters long seeds. Once hardened, elongated ripe pods split open on one side to reveal and release the flat reddish-brown seeds attached to thin silky white fibers between September and October (Edible Wild Food). To spread themselves, they fluff up to catch the wind and are carried away with their seeds.

What is its environment?

Milkweed grows in a variety of environments: farmlands, along roadsides and in ditches, open wetlands, dry sandy areas, short- and tall-grass prairie, agricultural areas, riverbanks, irrigation ditches, and arid valleys (Commission for Environmental Cooperation).

Common milkweed faces ecological challenges like habitat fragmentation and loss, climate change and human activity, such as herbicide application and deforestation (Commission for Environmental Cooperation; Geest et al.; Wells; Despland).

How did it adapt to its environment?

Common milkweed can be propagated by both its seeds and through cuttings of its deep rhizomes. It can propagate so quickly that it can be invasive. As an important food source for over 450 species of insects, common milkweed is well-known to be the exclusive food source of monarch caterpillars and butterflies who feed on some portions of the plant (nectar, sap, leaves and flowers) (Doonan).

Different species of milkweed contain varying levels of toxic cardiac glycoside compounds, which are secreted through a thick and white milky-like sap when the plant is injured. However, common milkweed contains less of these toxins than its southern counterparts. Some insects (e.g. monarch butterflies) feeding on the milkweed turn the toxic compounds into a method of defense against predators by storing them in their own bodies.

Common milkweed is toxic only in large quantities and if ingested. Symptoms include vomiting, stupor, weakness, spasms by ingesting other species; one would need careful identification. Nevertheless, milkweeds are very useful in the garden and in agriculture as they can drive away predatory and parasitoid insects.

Negative relationships between milkweed and other species

1- Milkweed is poisonous due to the cardenolide they contain. Therefore, it is treated as a noxious weed and farmers usually use herbicides to control its population (NatureWatch).

2- Insects play a big part in the pollination process; however, only big insects pollinate it while small insects get stuck (NatureWatch).

Positive relationships between milkweed and other species

1- Milkweed has a survival relationship with monarch butterflies as it is their primary food source. Adult monarch butterflies must lay their eggs directly on these plants, in order to provide the young caterpillars with the food they need (NatureWatch).

2- Milkweed also makes these butterflies taste unfavourable to other predators. Consequently, the monarch caterpillars carry the ingested toxins to adulthood and the natural cycle continues. For this reason, the survival of the monarch butterfly is intimately linked to milkweeds (NatureWatch).

3- A variety of insects need to survive off of milkweed (e.g. large milkweed bugs, small milkweed bugs, red milkweed beetle, swamp milkweed beetle, etc.) (Hadley).

Uses: Benefits and Disadvantages

Milkweed has always played a huge role in helping humanity as it was used in medicine to treat infections. Milkweed stems have been used by Indigenous peoples to make ropes and bowstrings while empty pods are used by modern-day educators for crafts.

The silky fluffs are "six times more buoyant than cork and five times as warm as wool," according to William Cullina, author of Growing and Propagating Wildflowers of the United States and Canada. Thanks to their buoyancy, they were used as lifejackets substitute stuffing during the WW2, and today they are commercially used for stuffing comforters and pillows (Canadian Wildlife Federation).

Its insulation qualities allowed the production of winter coats and warm clothing for warmth and moisture repellency in the textile industry and the production of cars and ambulances for sound reduction in the automotive industry. These fluffs are especially appealing as they are lightweight and require less fuel to transport than other heavier materials (Canadian Wildlife Federation).

Also recognized for its oil-absorbing and waterproof ability, milkweeds create an efficient and natural response to oil spills. The milky sap contains latex, which can cause an allergic reaction in some sensitive individuals and can be irritating to the skin (Canadian Wildlife Federation). The latex can be produced into rubber.

Various parts of milkweed can be used for food, although it is toxic if not correctly prepared and can lead to harmful health outcomes.

These benefits contribute to an increasing interest in milkweed crops by Quebec farms. To grow, milkweed requires moist, fine or medium sandy, clay-like, or rocky calcareous soils. Milkweeds are not tolerant of shadows as they require a lot of sunlight. Common milkweed and showy milkweed (A. speciosa) are spreading milkweed species that mainly spread through their rhizomes.

Common milkweed is edible only under certain circumstances. If done properly, boiling can eliminate the bitter taste and toxicity of the sap. Their heads can be fried in batter and eaten. To do so, in the early summer, flower buds can be harvested during the first seven weeks. They appear as unripe broccoli heads but taste the same as shoots. These milkweed flower buds are delicious in stir-fries, soups, rice casseroles, etc (Edible Wild Food).

Recipes

Common milkweed is reported to be part of the traditional diets of many groups of native North Americans. In a study focusing on the Myaamia, historically living in northern Indiana, it was found that their traditional cooking practices (which consists of repeated boiling with several changes of water) helped decrease the amount of toxic cardiac glycosides in the plant considerably. Moreover, it seems that the shoots contained slightly lower levels of toxin than other parts of the plant (Everest)

Tips on collecting milkweed

Beginners should forage common wildweed with caution and under supervision of a more experienced forager because many other highly toxic species of milkweed look like common milkweed (Brill).

Do not touch eyes when collecting milkweed or wear gloves
Seed pods must be no bigger than 3 inch to be eaten
.

What can be eaten?

Common milkweed is a seasonal produce and different parts can be eaten at different moments of the year. Young shoots (up to 6 inches tall) in mid-spring, flower buds with immature leaves from the tops of the plants in late spring, flowers in early summer, and very young, firm seed pods midsummer (Brill)

Preparing milkweed

To eliminate the slightly toxic and bitter sap, common milkweed must be boiled in 3 changes of water (except for mature flowers, which only need to be blanched once) (Brill).

It is recommended to only eat a small amount of milkweed the first time since some people may still not tolerate it well (Brill).

Preparing milkweed to get rid of the toxic sap (Brill)

1-Bring 2 pots of water (a large and medium one) to a boil over high heat.

2-Place the milkweed in the medium pot and let it boil for 1 min. Drain and discard the water.

3-Refill the medium pot with boiling water from the large pot.

4-Repeat step 2.

5-Repeat step 3 and boil milkweed for 15 minutes this time. Drain and try to get as much water out of the milkweed by pressing on it with a slotted spoon.

If you find that there is still a bitter taste, don't eat it. The plant that you have cooked may not be common milkweed but a toxic look-alike like dogbane or another species of milkweed (Bergo).

We have adapted the following recipes to use the safest method of preparation for milkweed.

Simple Milkweed Shoot Salad (Brill)

Ingredients:

12 cups milkweed shoots

4 cloves garlic, chopped

1 tablespoon olive oil

1 tablespoon tamari or soy sauce

1 tablespoon brown rice vinegar or red wine vinegar

Directions:

Step 1: Prepare milkweed shoots according to instructions on page 19.

Step 2: Toss garlic into a skillet with olive oil over medium heat and cook until lightly browned.

Step 3: Add the garlic to the drained milkweed shoots and season with tamari and rice vinegar.

Milkweed Mac'nCheese (Lord)

Ingredients:

2 cups milkweed flowers, diced and rinsed

3 cups milk

8 oz dry pasta

2 tablespoons flour

1 ½ cup shredded cheddar cheese

2 tablespoon butter

Salt and pepper bread crumbs

Directions:

Step 1: Preheat the oven to 350 degrees.

Step 2: Boil pasta in salty water until al dente and drain

Step 3: Prepare milkweed flowers according to page 19 but only boiling once should be enough.

Step 4: Pour both milkweed and pasta into a 8x8 baking dish.

Step 5: While pasta is cooking, make sauce:

a. Melt butter in a saucepan, add flour and cook for 3 minutes while stirring.

b. Add milk, salt and pepper.

c. Stirring frequently, bring to simmer then remove from heat.

d. Add cheddar and stir.

Step 6: Pour sauce over pasta and sprinkle with breadcrumbs.

Step 7: Bake for 20-30 minutes.

Ingredients:

Young milkweed leaves

Milkweed flowers

Salt

Ho Chunk Milkweed Soup (National Native Network)

Directions:

Step 1: Prepare milkweed according to the page 19.

Step 2: Boil for 30-40 minutes with salt, stirring occasionally.

Spaghetti with Milkweed Pods (Bergo)

Ingredients:

4 oz dried spaghetti

3 oz milkweed pods (no bigger than 3 in and cut in ½ if longer than 1 in)

½ tablespoon thinly sliced garlic

¼ cup cooking oil or lard

Kosher salt to taste

Crushed red pepper flakes to taste

Fresh torn basil to taste

2 cups chopped tomatoes

Extra virgin olive oil to taste, for seasoning

Grated parmesan (optional)

Directions:

Step 1: Prepare milkweed flowers according to page 19

Step 2: Meanwhile, cook garlic in cooking oil in a pan until light brown.

Step 3: Add milkweed pods and stir-fry until tender and bright green.

Step 4: Season with salt and pepper, add tomatoes, crushed red pepper and cook for a few more minutes.

Step 5: Boil pasta until al dente, drain and add it to the pan. Cook some more.

Step 6: Drizzle some olive oil to taste, sprinkle with basil and serve garnished with parmesan

How to Plant Milkweed Seeds

Milkweed seeds need to go through cold stratification. It is important to give the seeds an exposure to cold weather because it helps the seed's outer cast break or soften, and therefore germinate to grow as a plant. For this reason, milkweed is programmed to scatter its seeds late fall to give the seeds a chance for germination during the winter period. Therefore, it is best to follow the natural cycle of milkweed and start planting it during late fall (American Meadows; Howell).

"Stratification is when a seed is moistened, chilled or frozen and thawed, breaking down germination inhibitors on the seed coat, such as waxes, hormones, oils or heavy coats" (Howell).

Fall
(American Meadows; Howell)

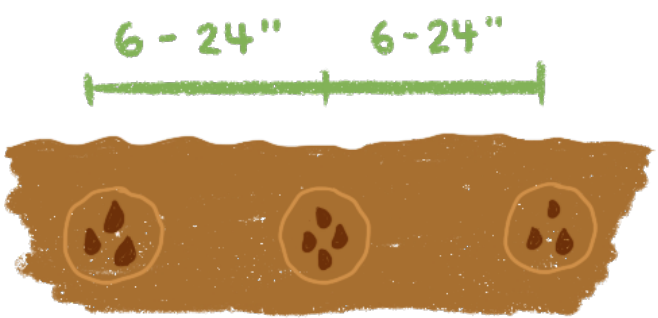
Depending on the time of planting, steps may differ as illustrated below:

1- In a sunny location, place the milkweed seeds in small groups (3 to 6) at a distance of 6 to 24 inches away from each other on the soil. The reason why it is better to keep them at a distance is because it is less likely to get insect or parasite infestation (American Meadows; Howell).

2- Pat the soil down then add another layer of soil above it.

3- Water the soil.

Since they are planted in fall, the seeds will be naturally stratified.



Spring
(American Meadows; Howell)

1- Put the seeds in damp sand or in a paper towel inside a zipped bag and put it in the refrigerator for 2 to 3 months. Make sure to store it in a place where the seeds are free from damage (e.g. fridge's inner corner)

2- After stratification, in a sunny location, place the milkweed seeds in small groups (3 to 6) at a distance of 6 in to 24 away from each other on the soil.

3- Pat the soil down then add another layer of soil above it.

4- Water the soil.

Since the seeds have been stratified in the refrigerator, it may take up to 40 days for the first sprouts to appear

Winter
(American Meadows; Howell)

It is recommended to initially plant the seeds indoors during winter.

1- Put the seeds in damp sand or a paper towel inside a zipped bag and put it in the refrigerator for 2 to 3 months. Make sure to store it in a place where the seeds are free from damage (e.g. fridge's inner corner).

2- After stratification, bring a pot with a seed-starting soil, and put 2 to 3 seeds inside.

3- Add a layer of about ¼ inch of this soil over the seeds.

4- Water it lightly and leave it under the sun.

5- When the ground is warm enough, and the plant has around 3 to 4 leaves or it reached around 3 inches tall, it should be transplanted.

6- Transplant the plant and its soil in an existing soil ground as it allows the plant to further grow and its roots can extend to a bigger surface.

Cold stratified seeds should germinate in around 10 days.

Upon transplanting, the plant might face a shock and the leaves might fall. Don't panic as this process naturally occurs to allow the plant to concentrate more on establishing its roots, then these leaves will grow back again.

For a better planting experience during winter or for those who want to start with a pod, using peat pots is the best way to go.

Instructions for peat pot planting (American Meadows):

1- The peat pot is drainable so make sure to put a pan or plate underneath it.

2- Put the seed-starting soil and water it a little bit.

3- Add 2-3 seeds then add another ¼ inch of the same soil.

4- Water the soil again lightly. If the soil is wet, don't add water until it feels dry.

5- Leave it in a sunny place.

6- For transplant, the peat pot can be placed as a whole in the soil (make sure no edges are above the ground) and the pot will break down over time. This process prevents the shock mentioned above where the leaves fall as the milkweed roots tend to be sensitive.

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